



PROJECT INFORMATION

Project and customer record

Project

PROJECT NAME Demo Humbucker	CREATED 21.09.2026
UPDATED 21.09.2026	PROJECT NOTES -

Customer

NAME Demo	ADDRESS -
PHONE -	EMAIL -
NOTES -	WIND DATE 21.09.2026



PROJECT INFORMATION

Pickup properties and test settings

Pickup

PICKUP ID Demo Humbucker	WIRE TYPE Plain Enamel
GAUGE AWG42	# WINDS 7000
POLARITY South	PHASE Negative
POLE INSULATOR Kapton Tape	PROTECTION Tissue
LEADS Waxed Pushback	START WIRE (HOT) Yellow
END WIRE (GROUND) Black	INDUCTANCE 5.4
DCR 7.9	

Pickup Selections

☒ GUITAR	☒ SINGLE COIL	☒ CLOCKWISE
☐ BASS	☐ HUMBUCKER	☐ COUNTERCLOCKWISE

Audio Routing

INPUT DEVICE M4	INPUT CHANNEL Channel 1
OUTPUT DEVICE Default - MacBook Pro Speakers	OUTPUT CHANNEL Channel 1 / Left

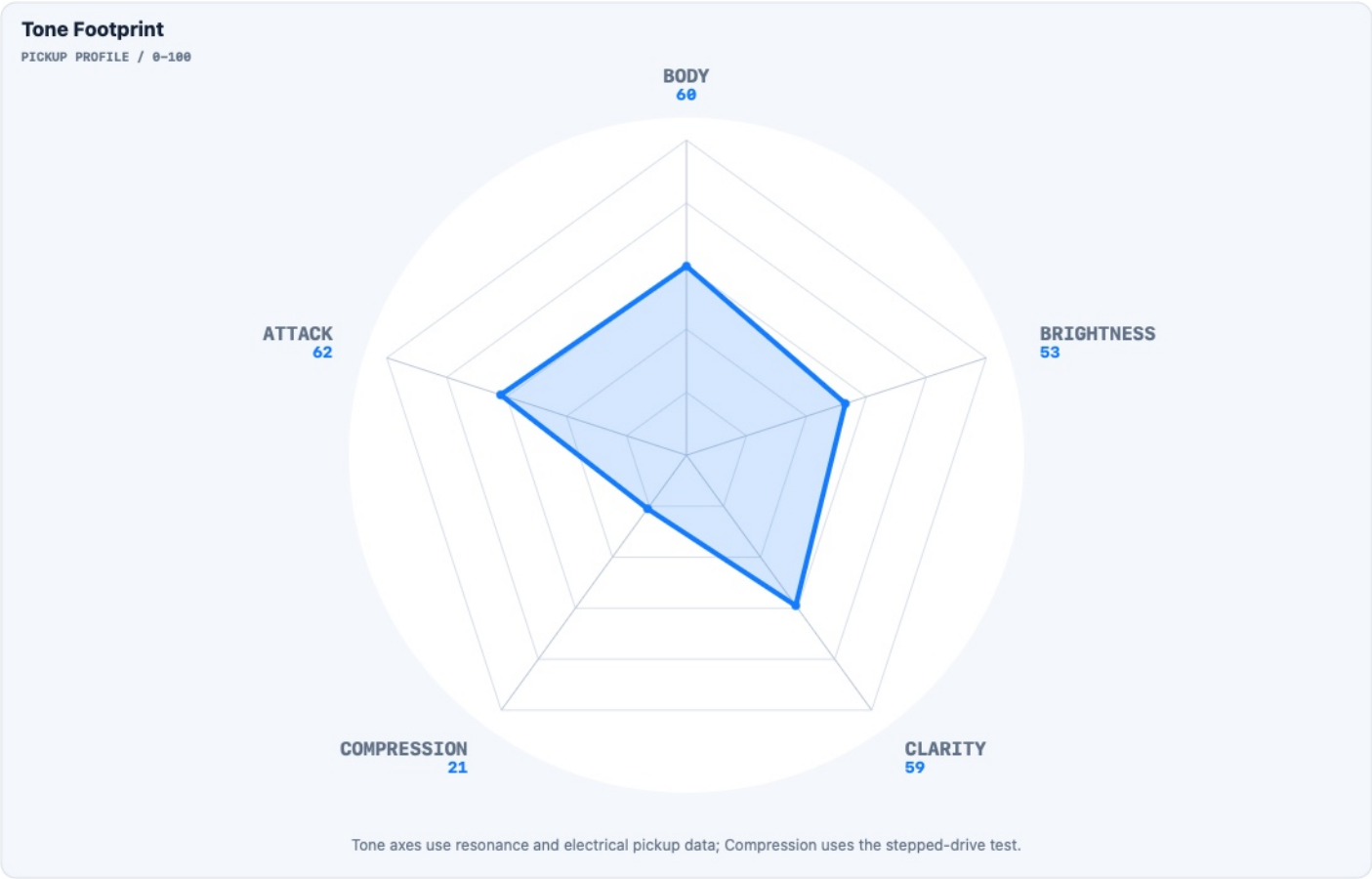
Measurement Settings

INPUT TRIM -7 dB	OUTPUT LEVEL -23 dBFS
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OVERVIEW

Tone footprint and key measurements



Key Measurements

FITTED RESONANCE 2.87 kHz	RELATIVE PHASE POSITIVE	DCR 7.9
INDUCTANCE 5.4	COMPRESSION 2.5 dB	

Tonal Characteristics

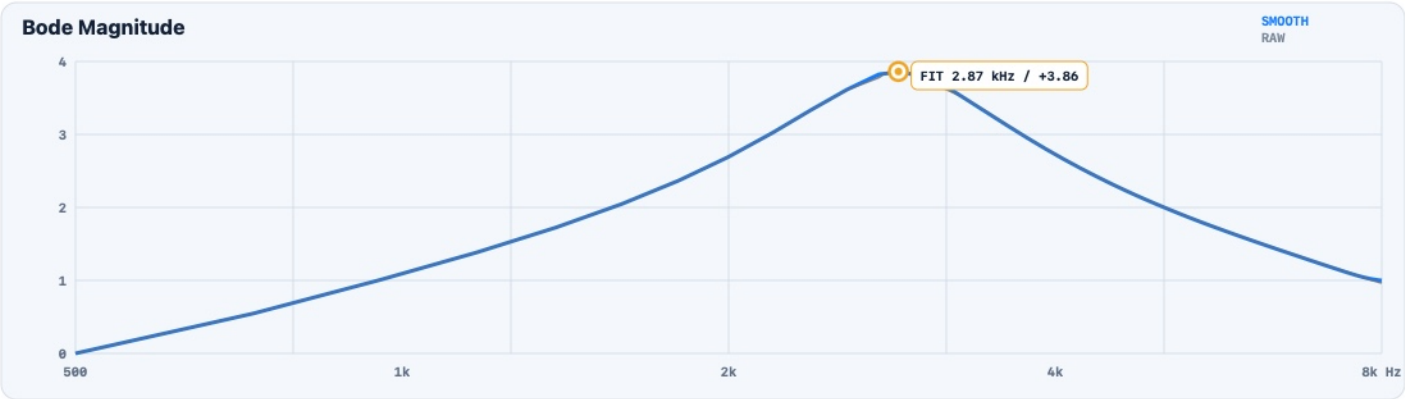
CALCULATED PROFILE

Warm / woody: Enhanced lower-mids, smooth treble, and a rounded note envelope. Effective profile 3.00 kHz from fitted resonance, inductance,



MEASUREMENTS

Bode response / Relative phase



Bode Response Data

FITTED RESONANCE 2.87 kHz	STRONGEST MEASURED BIN 2.85 kHz
FIT UNCERTAINTY ± 17 Hz	PEAK ABOVE MINIMUM +3.86
500 HZ TO 8 KHZ +0.97	VALID POINTS 69 / 69

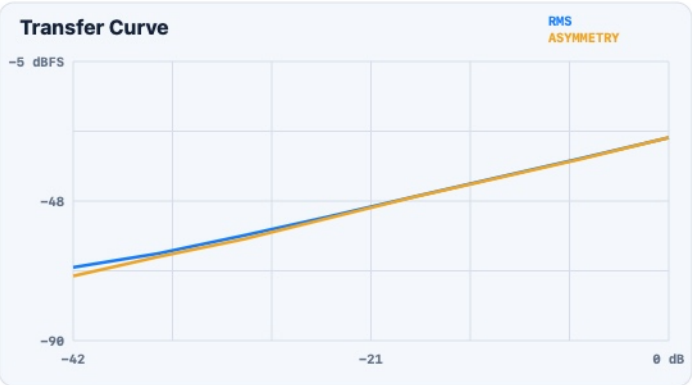
Relative Phase Data

RELATIVE PHASE POSITIVE	CONFIDENCE 100%
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MEASUREMENTS

Saturation / Spectrum



Saturation Data

EVEN THD AT MAXIMUM

-69.8 dB

COMPRESSION

2.5 dB

ODD THD AT MAXIMUM

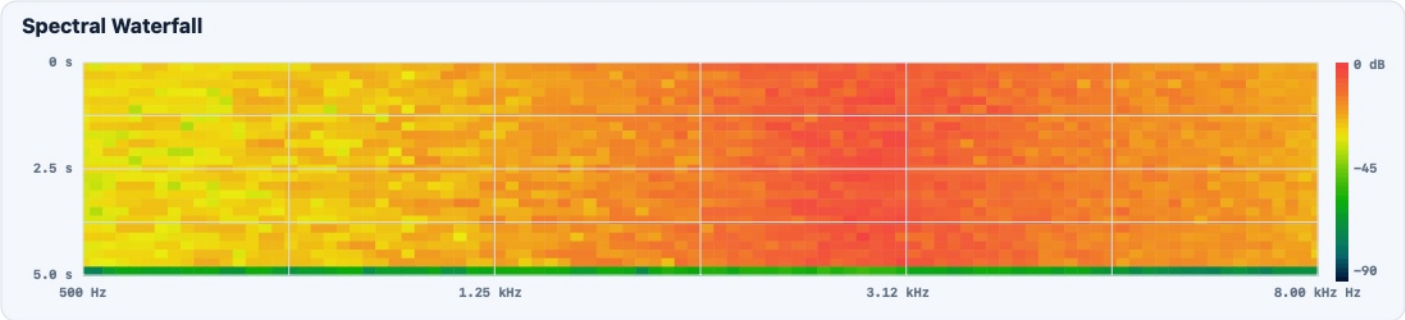
-67.1 dB

ASYMMETRY

0.0 dB

Spectrum Data

PEAK FREQUENCY	FUNDAMENTAL	RMS LEVEL	SAMPLE PEAK
2.90 kHz	1.38 kHz	-24.7 dBFS	-11.5 dBFS
SPECTRAL CENTROID	85% ROLL-OFF	SPECTRAL BANDWIDTH	SPECTRAL FLATNESS
3.32 kHz	4.29 kHz	1.17 kHz	45.2%
SPECTRAL SLOPE	THD ESTIMATE	EVEN / ODD	PEAK ASYMMETRY
-0.16 dB/oct	508.4%	+8.7 dB	-0.99 dB
TRANSIENT INDEX	LOW BAND	MID BAND	HIGH BAND
1.20x	-140.0 dB	-32.9 dB	-20.7 dB





Measurement glossary

Fitted resonance

Resonance estimated by a quadratic smooth-curve fit in log-frequency through the strongest local 5-7 direct points after adaptive peak refinement. It is not restricted to a measured frequency bin and includes the driver, fixture, pickup loading, cable, and interface.

Strongest measured bin

The actual direct analysis frequency with the highest robust-filtered response after the normal sweep and adaptive refinement. Compare it with the fitted resonance to see how far the smooth estimate moved between measured bins.

Fit uncertainty

A conservative plus/minus estimate combining half the closest measured-bin spacing with leave-one-out fit variation. It expresses local fit stability and frequency resolution, not calibrated laboratory metrology uncertainty.

Peak above minimum

Positive displayed difference between the fitted resonance peak and the lowest robust Bode point. The displayed minimum is fixed at 0.

500 Hz → 8 kHz

Displayed difference between the 8 kHz and 500 Hz endpoints. Positive is brighter across the measured span; negative indicates overall roll-off.

Valid points

Direct noise-cleared analysis points versus the normal sweep plus adaptive refinement total. Noise-limited gaps are interpolated for graph continuity but are not counted as direct points. Select 20, 30, 40, 50, or 60 normal points; closely spaced peak points are added automatically.

Peak frequency

The strongest averaged spectral component inside the selected span. It often marks the pickup/fixture resonance, but can also expose hum or a driver peak.

Fundamental

A harmonic-score estimate of the lowest periodic component supporting upper harmonics. Broadband noise may produce low confidence, so treat this as diagnostic rather than a guaranteed musical fundamental.

RMS level

Average signal energy in dBFS across the capture. It is the best single number for repeatable overall pickup output when gain and geometry stay fixed.

Sample peak

Largest absolute digital sample in dBFS. Values near 0 dBFS warn that the interface may be clipping even when RMS looks safe.

Spectral centroid

Power-weighted center of the spectrum. Higher values generally correspond to a brighter response; compare only with the same selected span.

85% roll-off

Frequency below which 85% of selected-band spectral power lies. It summarizes how far useful energy extends toward the high end.

Spectral bandwidth

Power-weighted standard deviation around the spectral centroid. Wider bandwidth means energy is distributed across a broader frequency range.

Spectral flatness

Geometric-to-arithmetic mean power ratio. Near 100% is noise-like and flat; low values indicate concentrated peaks or resonances.

Spectral slope

Linear trend of spectral level against log2 frequency, expressed in dB per octave. Negative values indicate high-frequency roll-off.

THD estimate

Energy at harmonics 2-8 relative to the estimated fundamental. Because white noise excites every bin, treat this as a comparative screening estimate rather than calibrated distortion.

Even / odd

Ratio of even-harmonic to odd-harmonic energy around the estimated fundamental. Positive dB means even harmonics dominate; use it only for like-for-like comparisons.

Peak asymmetry

Positive-to-negative waveform peak ratio in dB. A large magnitude may reflect magnetic/electronic asymmetry, DC bias, or clipping.



ANNEX

Measurement glossary

Transient index

Ratio between the strongest short-time RMS block and the median block RMS. It reveals unexpected knocks, handling noise, dropouts, or bursts during the noise run.

Low band

Integrated spectral level below 250 Hz, limited by the selected span. Useful for comparing hum, body, and low-frequency coupling.

Mid band

Integrated spectral level from 250 Hz to 2 kHz. This region contains much of the pickup's core musical output.

High band

Integrated spectral level above 2 kHz within the selected span. It tracks presence, resonance, and high-frequency loading.

Even THD at max

Combined 2nd, 4th, 6th, and 8th harmonics at maximum drive relative to the fundamental. Rapid even growth can indicate asymmetry, but the driver and interface can also create it.

Odd THD at max

Combined 3rd, 5th, and 7th harmonics at maximum drive relative to the fundamental. Odd growth often accompanies symmetric limiting or saturation.

Compression

Lost output growth between the lowest and highest drive steps compared with the commanded drive change. A positive value shows transfer-curve bending somewhere in the complete signal chain.

Asymmetry

Positive/negative half-cycle peak imbalance at maximum drive. Large values can reflect magnetic asymmetry, electronics, DC bias, or clipping.

Relative phase

Majority sign from five shaped-pulse responses. Positive or negative is relative to the unchanged driver face, winding, leads, and interface phase.

Confidence

Percentage of pulses agreeing with the majority sign. Below 80% is reported as indeterminate and should be repeated with better signal-to-noise.